Maths Exploring the Exercises at Terminal-02

Progressive exercises

E1 · Direct application. In a base of space, vectors $\vec{u}$ and $\vec{v}$ have for coordinates respective $(1;2;-1)$ and $(3;6;-3)$. Are they colinear? Justify.

E2 · Application and justification. In an orthonormal coordinate system of space, we consider $A(0;1;2)$ and $B(4;-1;0)$. Determine the coordinates of the midpoint of the $[AB]$then calculate the distance AB.

E3 · Method to be organised. In a base of space, we consider $\vec{u} = (1;0;1)$, $\vec{v} = (0;1;1)$ and $\vec{w} = (2;3;5)$. Express $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$, and then deduce that these three vectors are coplanar.

E4 · Choice of strategy. In a base of space, vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ have for coordinates $(1;0;1)$, $(0;1;1)$ and $(1;1;0)$. The triplet $(\vec{u},\vec{v},\vec{w})$ Is it a base of space? Justify.

E5 · Synthesis or demonstration. In a space coordinate system, we consider $A(1;0;2)$, $B(3;-1;4)$ and $C(5;-2;6)$. Determine whether C belongs to the right (AB). Justify by vector calculation.